



Zagazig University

Faculty of Engineering



ECE 321: Electronic Circuits

Lecture (05) & (06)

Dr. Ahmed Wahba

Assistant Professor, Electronics and Communications Engineering Department, Faculty of Engineering, Zagazig University, Zagazig, Egypt.

Ph.D, Department of Electronics Science and Technology, University of Science and Technology of China (USTC), Hefei, China.

**ECE 321 Electronic Circuits, ECE dept., Faculty of Engineering, Zagazig University, Fall 2022,
by Dr. Ahmed Wahba**

Outline

- ☐ **Solved Problems on Current Mirrors**
- ☐ **Differential Amplifier**
 - ☐ **Amplifier Types**
 - ☐ **Differential Vs Single-ended Signaling**
 - ☐ **Common-mode and differential signals**
 - ☐ **BJT Differential Amplifier**
 - ☐ **Large Signal Operation of BJT Differential Amplifier**

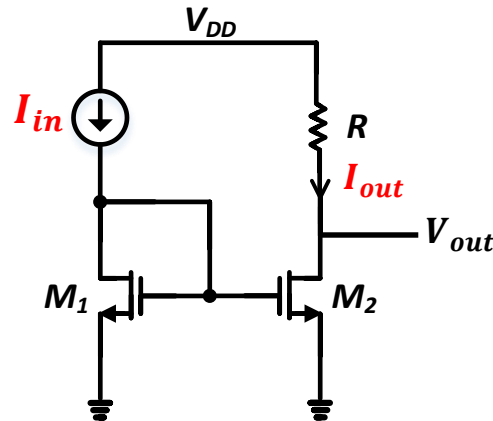
Outline

- ❑ **Solved Problems on Current Mirrors**
- ❑ **Differential Amplifier**
 - ❑ **Amplifier Types**
 - ❑ **Differential Vs Single-ended Signaling**
 - ❑ **Common-mode and differential signals**
 - ❑ **BJT Differential Amplifier**
 - ❑ **Large Signal Operation of BJT Differential Amplifier**

Problem # 1 (1/3)

□ For the shown current mirror circuit,

- Find $\left(\frac{I_{out}}{I_{in}}\right)$ and $V_{out,min}$ and discuss how can you size the transistors to achieve better performance.
- Plot I_{out} and V_{out} versus R as R changes from 0 to ∞ . $V_{DD} = 2.5$ V, $I_{in} = 1$ mA, $(W/L)_2 = 2(W/L)_1 = 10$, $\mu_n C_{ox} = 0.5$ mA/V², $V_{tn} = 0.4$ V, and $\lambda = 0$.



Solution

- $I_G = 0$, $V_{GS1} = V_{GS2} = V_{GS} \rightarrow V_{ov1} = V_{ov2} = V_{ov}$
- $I_{in} = I_{D1} = \frac{1}{2} \mu_n C_{ox} (W/L)_1 (V_{GS1} - V_{tn})^2 (1 + \lambda V_{DS1})$
- $I_{out} = I_{D2} = \frac{1}{2} \mu_n C_{ox} (W/L)_2 (V_{GS2} - V_{tn})^2 (1 + \lambda V_{DS2})$

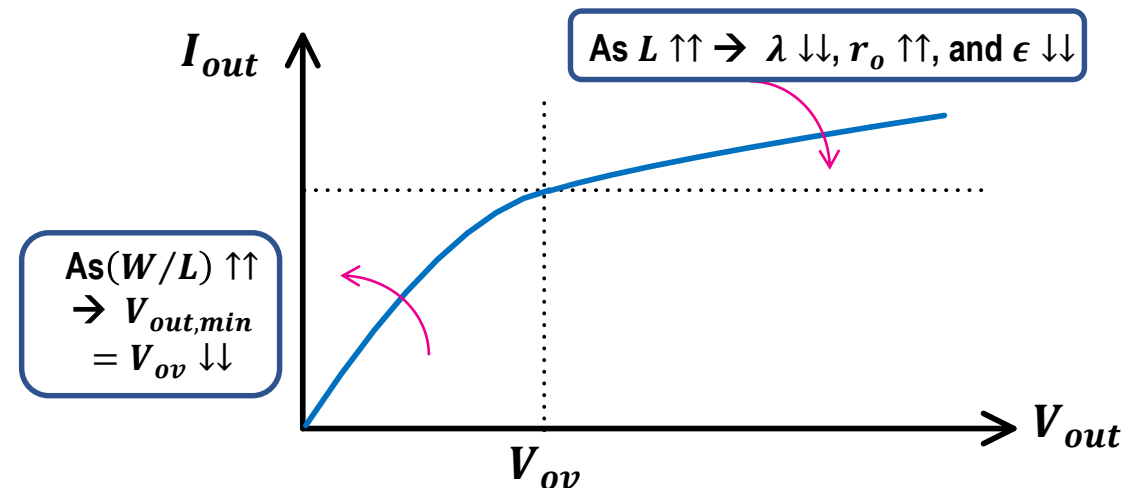
$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times [1 + \lambda (V_{DS2} - V_{DS1})] = \frac{(W/L)_2}{(W/L)_1} \times [1 + \epsilon]$$

$$\square V_{out,min} = V_{ov2} = \sqrt{\frac{2 I_{out}}{\beta_n}} = \sqrt{\frac{2 I_{out}}{\mu_n C_{ox} (W/L)_2}}$$

□ For better performance:

□ $V_{out,min} \downarrow \downarrow$: High (W/L) is required

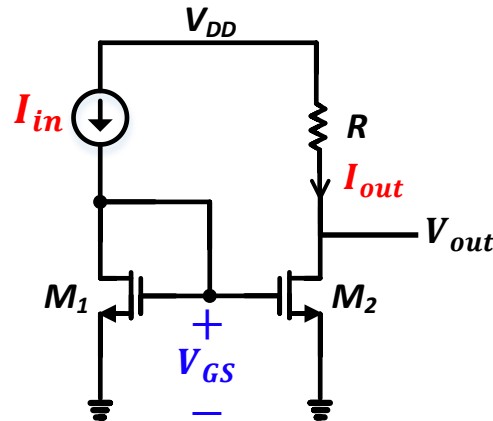
□ $\epsilon \downarrow \downarrow$: $\lambda \downarrow \downarrow$, as $\lambda \propto \frac{1}{L}$, High (L) is required.



Problem # 1 (2/3)

□ For the shown current mirror circuit,

- Find $\left(\frac{I_{out}}{I_{in}}\right)$ and $V_{out,min}$ and discuss how can you size the transistors to achieve better performance.
- Plot I_{out} and V_{out} versus R as R changes from 0 to ∞ . $V_{DD} = 2.5$ V, $I_{in} = 1$ mA, $(W/L)_2 = 2(W/L)_1 = 10$, $\mu_n C_{ox} = 0.5$ mA/V², $V_{tn} = 0.4$ V, and $\lambda = 0$.



Solution

- As $\lambda = 0 \rightarrow \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} = 2$
- $I_{out} = 2I_{in} = 2$ mA As long as M2 operates in saturation region

□ As R increases, V_{out} will decrease $\rightarrow$ pushing M2 towards triode region. So, Get $V_{ov} = V_{GS} - V_{tn}$ to check

□ $\beta_{n1} = \mu_n C_{ox} (W/L)_1 = 0.5 \text{ mA/V}^2 \times 5 = 2.5 \text{ mA/V}^2$

□ $\beta_{n2} = \mu_n C_{ox} (W/L)_2 = 0.5 \text{ mA/V}^2 \times 10 = 5 \text{ mA/V}^2$

□ Get V_{ov}

□ $I_{in} = \frac{\beta_{n1}}{2} V_{ov}^2$

□ $1 \text{ mA} = 1.25 \text{ mA/V}^2 \times V_{ov}^2$

□ $V_{ov} = 0.894 \text{ V}$

□ $V_{GS} = V_{ov} + V_{tn} = 1.294 \text{ V}$

□ Find R that makes M2 at the edge of saturation

□ $V_{out} = V_{ov} = 0.894 \text{ V}$

□ $R = \frac{V_{DD} - V_{ov}}{I_{out}} = \frac{2.5 \text{ V} - 0.894 \text{ V}}{2 \text{ mA}} = 803 \Omega$

□ M2 operating regions

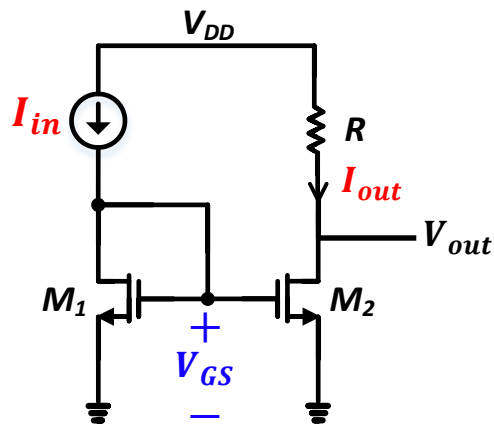
□ Saturation: $0 < R < 803 \Omega$

□ Triode: $R > 803 \Omega$

problem # 1 (3/3)

□ For the shown current mirror circuit,

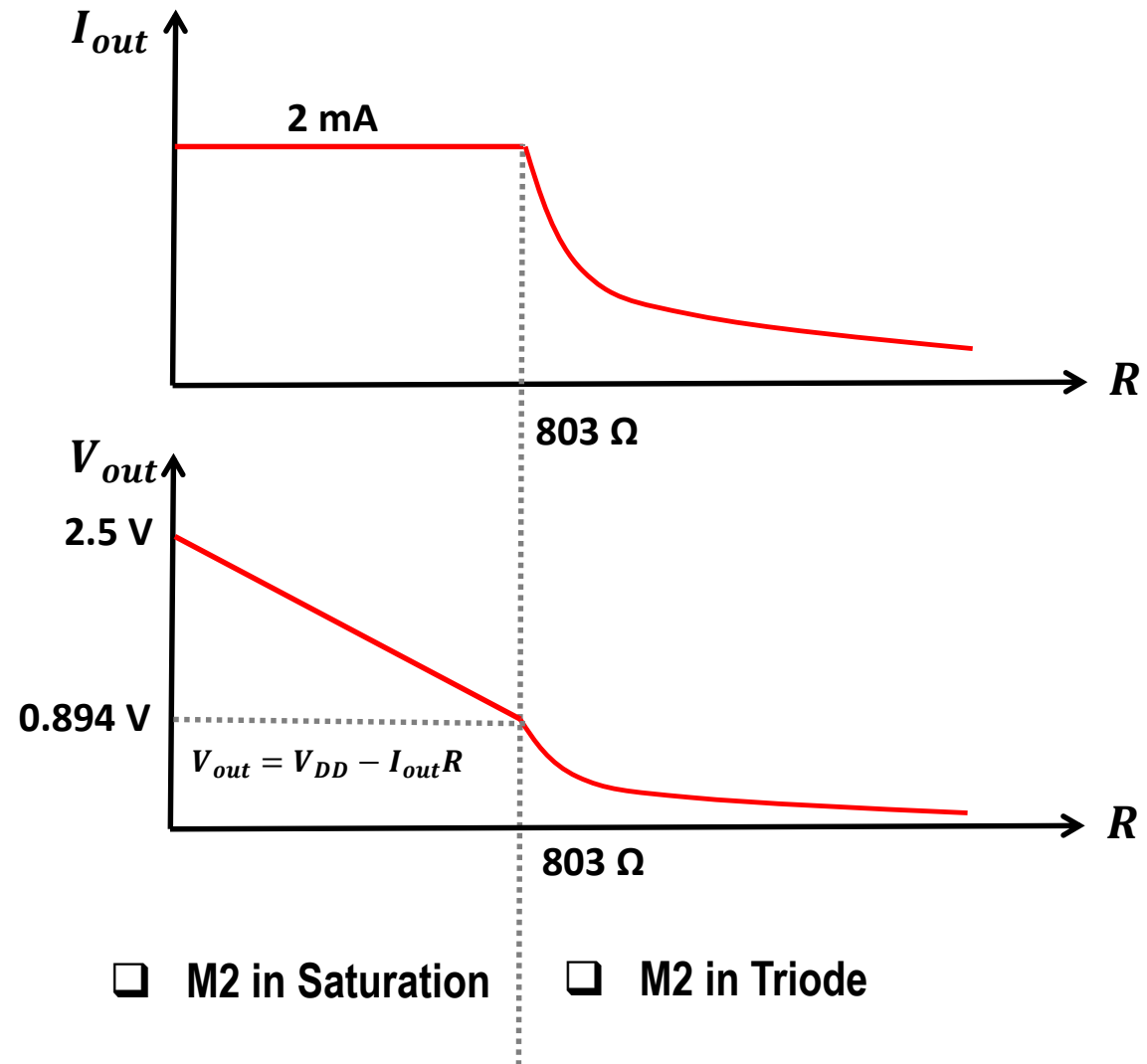
- Find $\left(\frac{I_{out}}{I_{in}}\right)$ and $V_{out,min}$ and discuss how can you size the transistors to achieve better performance.
- Plot I_{out} and V_{out} versus R as R changes from 0 to ∞ . $V_{DD} = 2.5$ V, $I_{in} = 1$ mA, $(W/L)_2 = 2(W/L)_1 = 10$, $\mu_n C_{ox} = 0.5$ mA/V², $V_{tn} = 0.4$ V, and $\lambda = 0$.



Solution

□ M2 operating regions

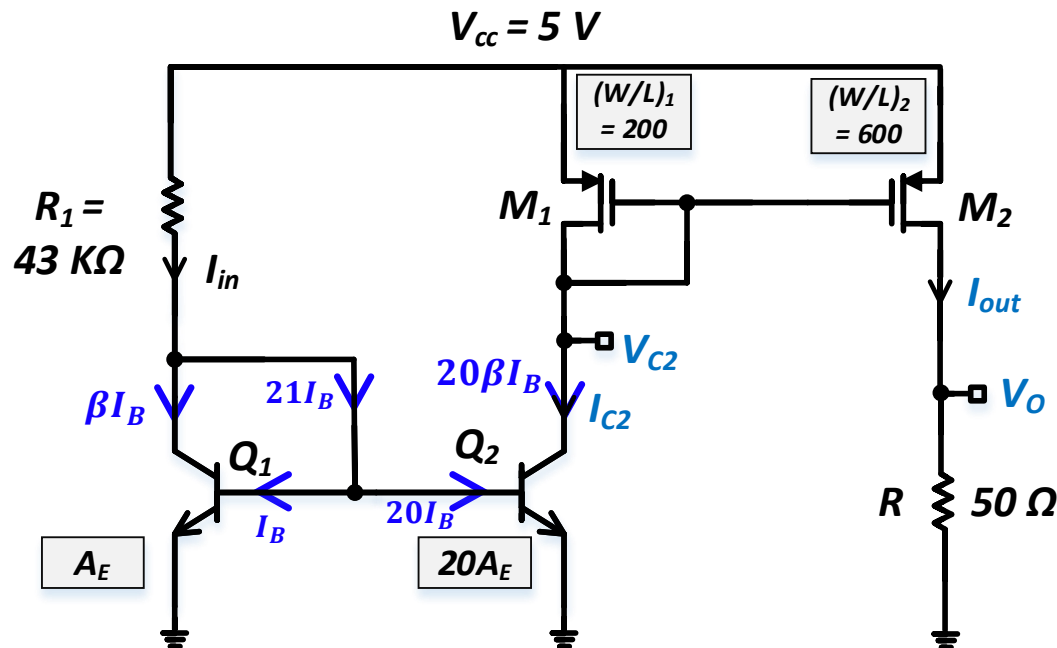
- Saturation: $0 < R < 803\Omega$
- Triode: $R > 803\Omega$



- M2 in Saturation
- M2 in Triode

Problem # 2 (1/3)

- For the shown current mirror circuit,
 - Find I_{C1} , V_{C2} , I_{out} , V_o . The npn transistors have $\beta = 120$. The PMOS transistors have $\mu_p C_{ox} = 0.2 \text{ mA/V}^2$, $V_{tp} = -0.4 \text{ V}$, and $\lambda_p = 0.1 \text{ V}^{-1}$.
 - If R is increased to 800Ω , find the percentage change in I_{out}

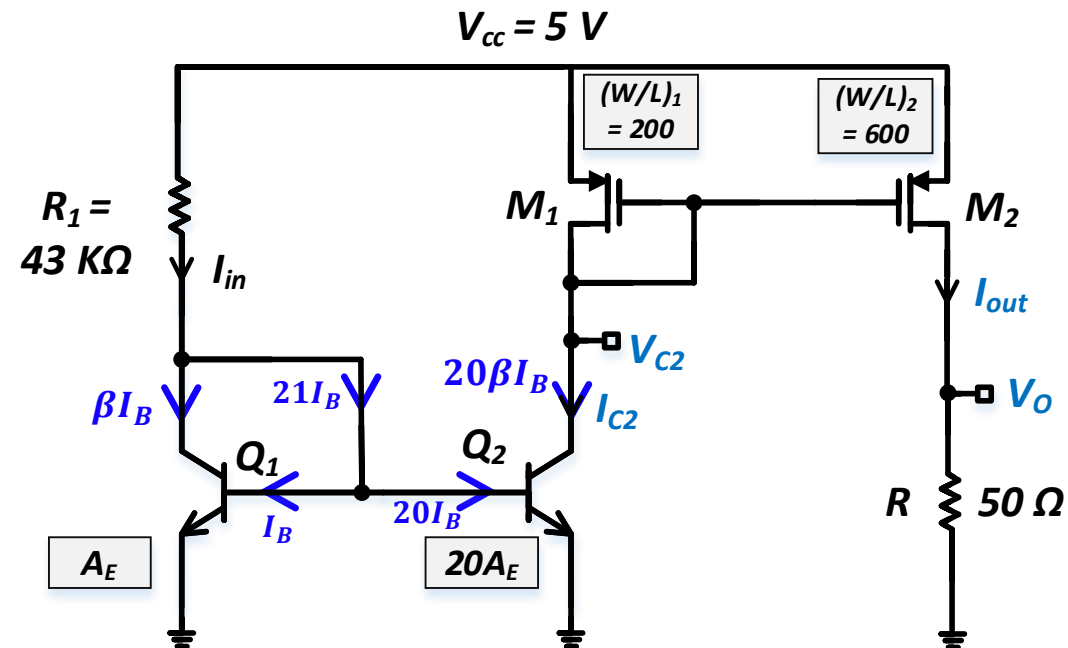


Solution

- **Get I_{in}**
 - $I_{in} = \frac{V_{CC} - V_{BE1}}{R_1} = \frac{5 \text{ V} - 0.7 \text{ V}}{43 \text{ K}} = 0.1 \text{ mA}$
- **Get I_{C2}**
 - $I_{in} = I_B(\beta + 21)$ □ $I_{C2} = 20\beta I_B$
 - $\frac{I_{C2}}{I_{in}} = \frac{20\beta}{\beta + 21}$ □ $I_{C2} = I_{in} \frac{20\beta}{\beta + 21} = 0.1 \text{ mA} \times \frac{20 \times 120}{120 + 21} = 1.702 \text{ mA}$
- **Get V_{C2}**
 - $I_{C2} = I_{SD(M_1)} = \frac{\beta_{p1}}{2} (|V_{GS1}| - |V_{tp}|)^2 (1 + \lambda_p |V_{DS1}|)$
 - $\beta_{p1} = \mu_p C_{ox} (W/L)_1 = 0.2 \text{ mA/V}^2 \times 200 = 40 \text{ mA/V}^2$
 - Note that: $V_{DS1} = V_{GS1}$
 - $0.02 (|V_{GS1}| - 0.4)^2 (1 + 0.1 |V_{GS1}|) = 1.702 \text{ mA}$
 - Solve 3rd order equation:
 - $|V_{GS1}| = -9.99 \text{ V}$ □ $|V_{GS1}| = 0.68226 \text{ V}$ □ $|V_{GS1}| = 0.11 \text{ V}$
 - Accepted

Problem # 2 (2/3)

- For the shown current mirror circuit,
 - Find I_{C1} , V_{C2} , I_{out} , V_o . The npn transistors have $\beta = 120$. The PMOS transistors have $\mu_p C_{ox} = 0.2 \text{ mA/V}^2$, $V_{tp} = -0.4 \text{ V}$, and $\lambda_p = 0.1 \text{ V}^{-1}$.
 - If R is increased to 800Ω , find the percentage change in I_{out}

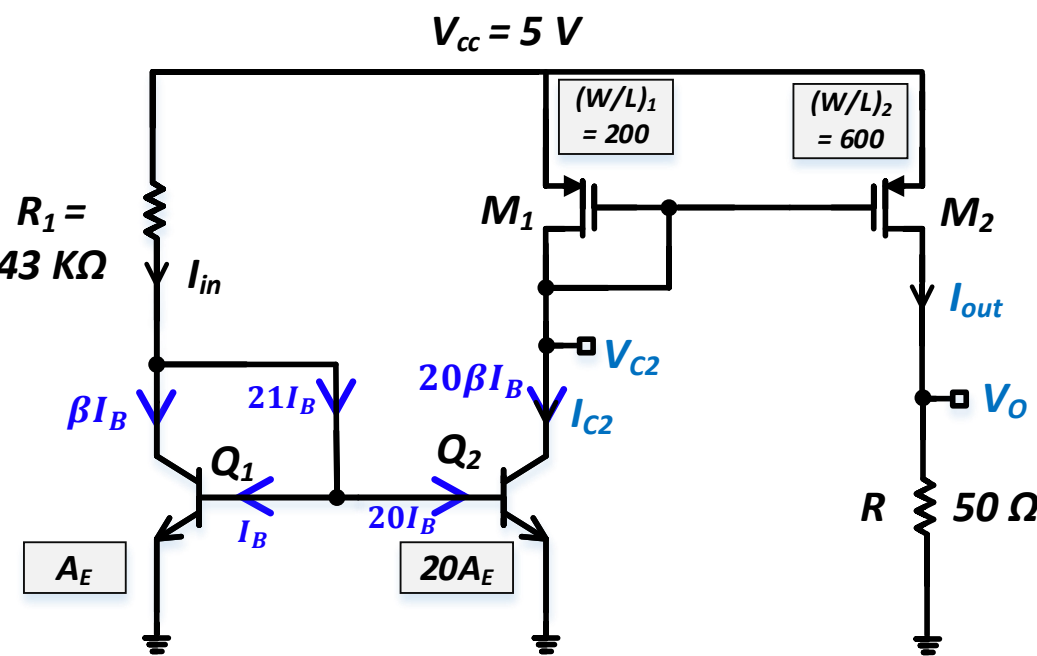


Solution

- $|V_{GS1}| = 0.6823 \text{ V}$ □ $V_{C2} = V_{DD} - |V_{GS1}| = 4.3177 \text{ V}$
- **Get I_{out} and V_{out}**
 - M1 and M2 are current mirror $\rightarrow V_{GS1} = V_{GS2}$
 - $I_{SD2} = I_{out} = \frac{\beta_{p2}}{2} (|V_{GS2}| - |V_{tp}|)^2 (1 + \lambda_p |V_{DS2}|)$
 - $\beta_{p2} = \mu_p C_{ox} (W/L)_2 = 0.2 \text{ mA/V}^2 \times 600 = 0.12 \text{ A/V}^2$
 - $|V_{GS2}| = |V_{GS1}| = 0.6823 \text{ V}$
 - $|V_{DS2}| = V_{CC} - I_{out} R$
 - $I_{out} = 0.06(0.6823 - 0.4)^2 (1 + 0.1[5 - 50I_{out}])$
 - $I_{out} = 7.005 \text{ mA}$
 - $V_{out} = I_{out} R = 0.35 \text{ V}$

Problem # 2 (3/3)

- For the shown current mirror circuit,
 - Find I_{C1} , V_{C2} , I_{out} , V_o . The npn transistors have $\beta = 120$. The PMOS transistors have $\mu_p C_{ox} = 0.2 \text{ mA/V}^2$, $V_{tp} = -0.4 \text{ V}$, and $\lambda_p = 0.1 \text{ V}^{-1}$.
 - If R is increased to 800Ω , find the percentage change in I_{out}

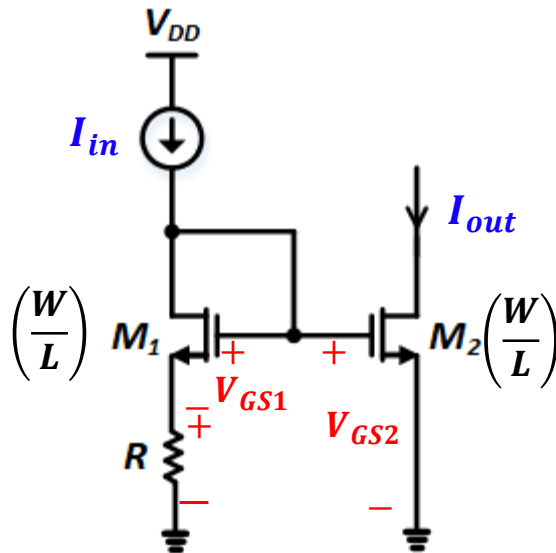


Solution

- If R is increased to 800Ω , find the percentage change in I_{out}
 - $I_{out} = 0.06(0.6823 - 0.4)^2(1 + 0.1[5 - 800I_{out}])$
 - $I_{out} = 5.188 \text{ mA}$ □ $V_{out} = I_{out} R = 4.15 \text{ V}$
 - $|V_{DS2}| = V_{CC} - V_{out} = 0.85 \text{ V}$ Check in saturation (OK)
- @ $R = 50 \Omega \rightarrow I_{out} = 7.005 \text{ mA}$
- @ $R = 800 \Omega \rightarrow I_{out} = 5.188 \text{ mA}$
- % change = $\frac{5.188 \text{ mA} - 7.005 \text{ mA}}{7.005 \text{ mA}} \times 100 = -25.9 \%$

Problem # 3

- For the shown current mirror circuit,
- Find R in terms of I_{in} such that $I_{out} = 2I_{in}$



Solution

- $I_G = 0$
- $I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_{tn})^2$
- $V_{GS} = \sqrt{\frac{2 I_D}{\mu_n C_{ox} (W/L)}} + V_{tn}$

$$\square V_{GS2} = V_{GS1} + I_{in} R \quad (1)$$

$$\square V_{GS1} = \sqrt{\frac{2 I_{in}}{\mu_n C_{ox} (W/L)}} + V_{tn} \quad (2)$$

$$\square V_{GS2} = \sqrt{\frac{2 I_{out}}{\mu_n C_{ox} (W/L)}} + V_{tn} = \sqrt{\frac{4 I_{in}}{\mu_n C_{ox} (W/L)}} + V_{tn} \quad (3)$$

- Sub. With (2) and (3) into (2)

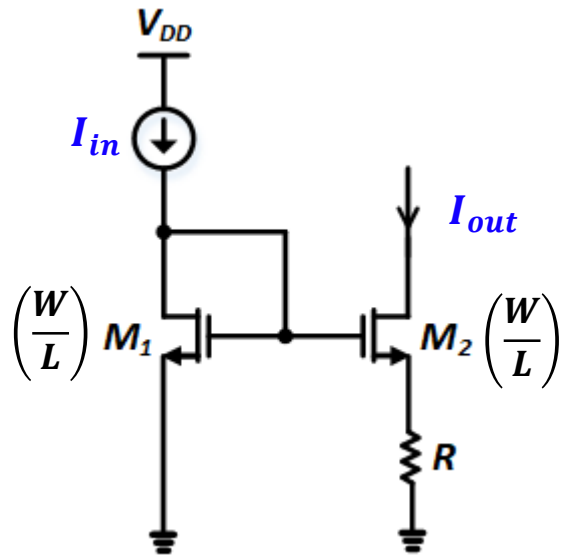
$$\square \sqrt{\frac{4 I_{in}}{\mu_n C_{ox} (W/L)}} + V_{tn} = \sqrt{\frac{2 I_{in}}{\mu_n C_{ox} (W/L)}} + V_{tn} + I_{in} R$$

$$\square \sqrt{\frac{I_{in}}{\mu_n C_{ox} (W/L)}} [2 - \sqrt{2}] = I_{in} R$$

$$\square R = \frac{[2 - \sqrt{2}]}{\sqrt{I_{in} \mu_n C_{ox} (W/L)}}$$

Problem # 4 & 5

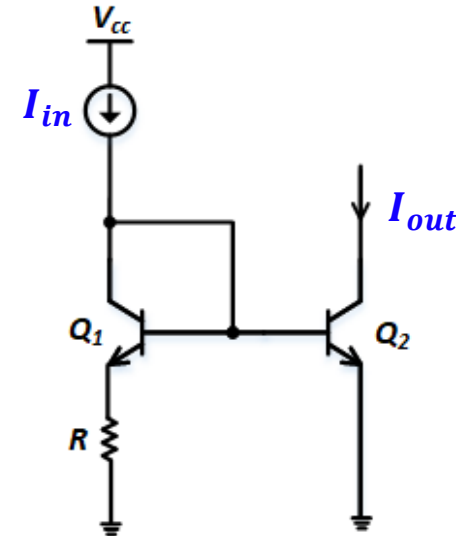
- For the shown current mirror circuit,
 □ Find R in terms of I_{in} such that $I_{out} = 0.5 I_{in}$



Solution

$$\square R = \frac{[2\sqrt{2}-2]}{\sqrt{I_{in} \mu_n C_{ox} (W/L)}}$$

- For the shown current mirror circuit,
 □ Find R in terms of I_{in} such that $I_{out} = 2 I_{in}$. Q1 and Q2 are matched. Assume $\beta = \infty$.

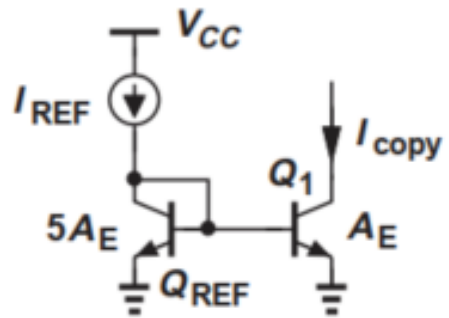


Solution

$$\square R = \frac{V_T \ln(2)}{I_{in}}$$

Problem # 6

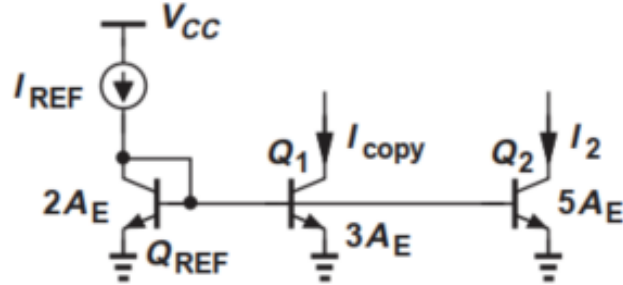
□ For each of the shown circuits, find $\frac{I_{copy}}{I_{REF}}$ and the systematic gain error.



□ $\frac{I_{copy}}{I_{REF}} = \frac{\beta}{5\beta+6}$

□ $\frac{I_{copy}}{I_{REF}} = \frac{1}{5} \left[1 - \frac{1.2}{\beta+1.2} \right]$

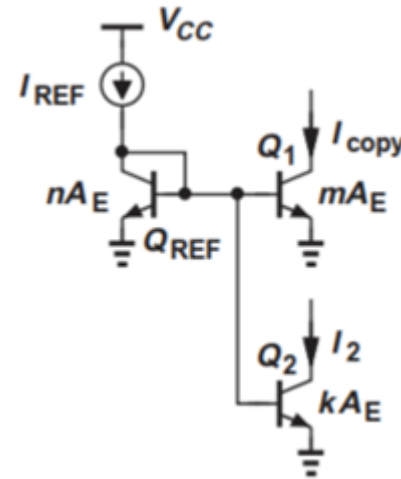
□ $\epsilon = \frac{1.2}{\beta+1.2}$



□ $\frac{I_{copy}}{I_{REF}} = \frac{3\beta}{2\beta+10}$

□ $\frac{I_{copy}}{I_{REF}} = \frac{3}{2} \left[1 - \frac{5}{\beta+5} \right]$

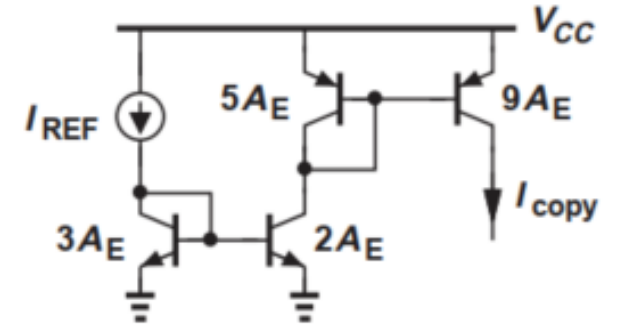
□ $\epsilon = \frac{5}{\beta+5}$



□ $\frac{I_{copy}}{I_{REF}} = \frac{m\beta}{n\beta+n+m+k}$

□ $\frac{I_{copy}}{I_{REF}} = \frac{m}{n} \left[1 - \frac{1+\left(\frac{m+k}{n}\right)}{\beta+1+\left(\frac{m+k}{n}\right)} \right]$

□ $\epsilon = \frac{1+\left(\frac{m+k}{n}\right)}{\beta+1+\left(\frac{m+k}{n}\right)}$



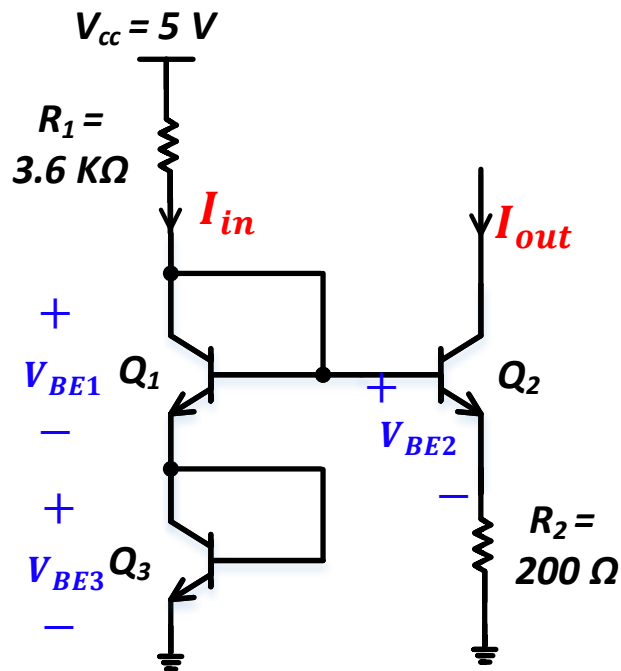
□ $\frac{I_{copy}}{I_{REF}} = \left(\frac{2\beta}{3\beta+5} \right) \left(\frac{9\beta}{5\beta+14} \right)$

□ $\frac{I_{copy}}{I_{REF}} = \frac{6}{5} \left[1 - \frac{67\beta+70}{15\beta^2+67\beta+70} \right]$

□ $\epsilon = \frac{67\beta+70}{15\beta^2+67\beta+70}$

Problem # 7 (1/2)

- For each of the shown current mirror circuit, Q_1 , Q_2 , and Q_3 are matched with $I_S = 5 \times 10^{-15} \text{ A}$ and $V_A = 120 \text{ V}$. Assume that $\beta = \infty$. Find I_{out} , V_{IN} , $V_{out,min}$, and R_{out}
- Find R_2 such that $I_{out} = 0.5 \text{ mA}$, and find the corresponding R_{out}



Solution

□ As $\beta = \infty$, $I_B = 0$, $I_C = I_E$ □ $I_{in} = I_{C1} = I_{C3}$ and $I_{out} = I_{C2}$

□ $I_{in} = \frac{V_{CC} - (V_{BE1} + V_{BE3})}{R_1} = \frac{5 \text{ V} - 1.4 \text{ V}}{3.6 \text{ K}\Omega} = 1 \text{ mA}$

□ $V_{BE1} + V_{BE3} = V_{BE2} + I_{out}R_2$ □ $V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$

□ $V_T \ln\left(\frac{I_{in}}{I_S}\right) + V_T \ln\left(\frac{I_{in}}{I_S}\right) = V_T \ln\left(\frac{I_{out}}{I_S}\right) + I_{out}R_2$

□ $V_T \ln\left(\frac{I_{in}^2}{I_{out} \times I_S}\right) = I_{out}R_2$ □ $I_{out} = \frac{V_T}{R_2} \ln\left(\frac{I_{in}^2}{I_{out} \times I_S}\right)$

□ Get I_{out} using trial and error □ $I_{out} = 3.23 \text{ mA}$

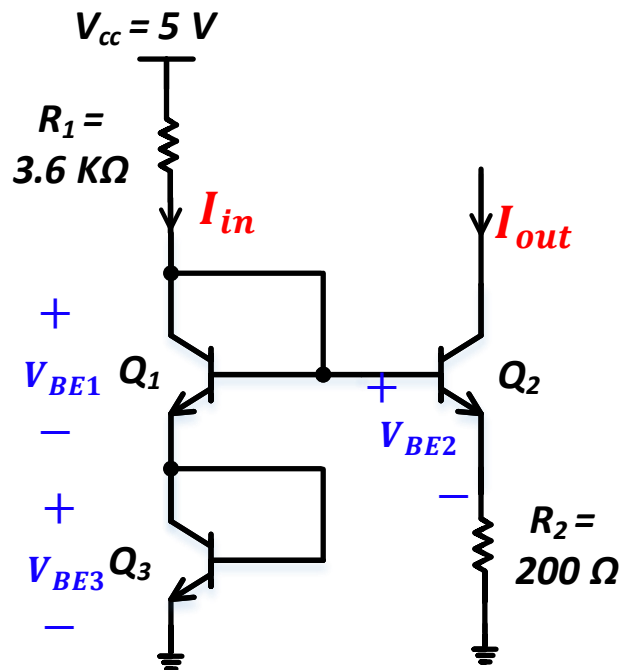
□ $V_{IN} = 1.4 \text{ V}$

□ $V_{out,min} = I_{out}R_2 + V_{CE2,sat} = (3.23 \text{ mA} \times 200\Omega) + 0.2 = 0.846 \text{ V}$

Problem # 7 (2/2)

□ For each of the shown current mirror circuit, Q_1 , Q_2 , and Q_3 are matched with $I_S = 5 \times 10^{-15} \text{ A}$ and $V_A = 120 \text{ V}$. Assume that $\beta = \infty$. Find I_{out} , V_{IN} , $V_{out,min}$, and R_{out}

□ Find R_2 such that $I_{out} = 0.5 \text{ mA}$, and find the corresponding R_{out}



Solution

$$\square R_{out} \approx r_{o2}[1 + g_{m2}(R_2 \parallel r_{\pi2})] \approx r_{o2}[1 + g_{m2} R_2]$$

$$\square r_{o2} = \frac{V_A}{I_{out}} = \frac{120 \text{ V}}{3.23 \text{ mA}} = 37.15 \text{ K}\Omega.$$

$$\square g_{m2} = \frac{I_{out}}{V_T} = \frac{3.23 \text{ mA}}{26 \text{ mV}} = 0.124 \text{ S}$$

$$\square R_{out} \approx 958.47 \text{ K}\Omega$$

$$\square r_{\pi2} = \beta r_{e2} = \infty$$

$$\square I_{out} = \frac{V_T}{R_2} \ln \left(\frac{I_{in}^2}{I_{out} \times I_S} \right)$$

$$\square I_{in} = 1 \text{ mA}, I_{out} = 0.5 \text{ mA} \quad \text{Solve \& get } R_2$$

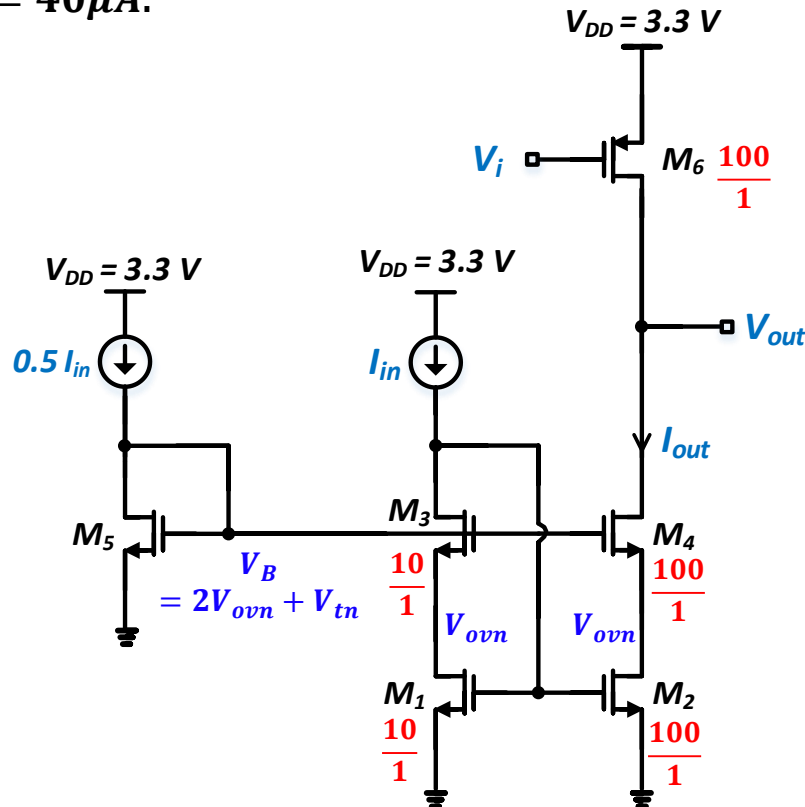
$$\square R_2 = \frac{V_T}{I_{out}} \ln \left(\frac{I_{in}^2}{I_{out} \times I_S} \right)$$

$$\square R_2 = 1.389 \text{ K}\Omega$$

$$\square R_{out} \approx r_{o2}[1 + g_{m2} R_2] = 6.65 \text{ M}\Omega$$

Problem # 8 (1/3)

- The shown amplifier uses the modified cascode current mirror to boost its gain. $\mu_n C_{ox} = 100 \mu A/V^2$, $\mu_p C_{ox} = 50 \mu A/V^2$, $V_{tn} = |V_{tp}| = 0.4 V$, $\lambda_n = |\lambda_p| = 0.1 V^{-1}$. (a) if $I_{in} = 20 \mu A$, find $(W/L)_5$ for maximum output swing, find A_v , $V_{out,min}$, and $V_{out,max}$ (b) Repeat (a) if $I_{in} = 40 \mu A$.



Solution

- For $\epsilon = 0$, $\rightarrow V_{DS1} = V_{DS2}$ and for maximum output swing, $V_{out,min}$ must be as small as possible ($2V_{ovn}$). $\rightarrow V_B = 2V_{ovn} + V_{tn}$

□ $\frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} = 10$ □ @ $I_{in} = 20 \mu A \rightarrow I_{out} = 200 \mu A$

- Note: $V_{ov1} = V_{ov2} = V_{ov3} = V_{ov4} = V_{ovn}$

- Get V_{ovn}

□ $V_{ovn} = \sqrt{\frac{2 I_{in}}{\mu_n C_{ox} (W/L)_1}} = \sqrt{\frac{2 \times 20 \mu A}{100 \mu A/V^2 \times 10}} = 0.2 V$

- Get $(W/L)_5$ such that $V_B = V_{GS5} = 2V_{ovn} + V_{tn} = 0.8 V$

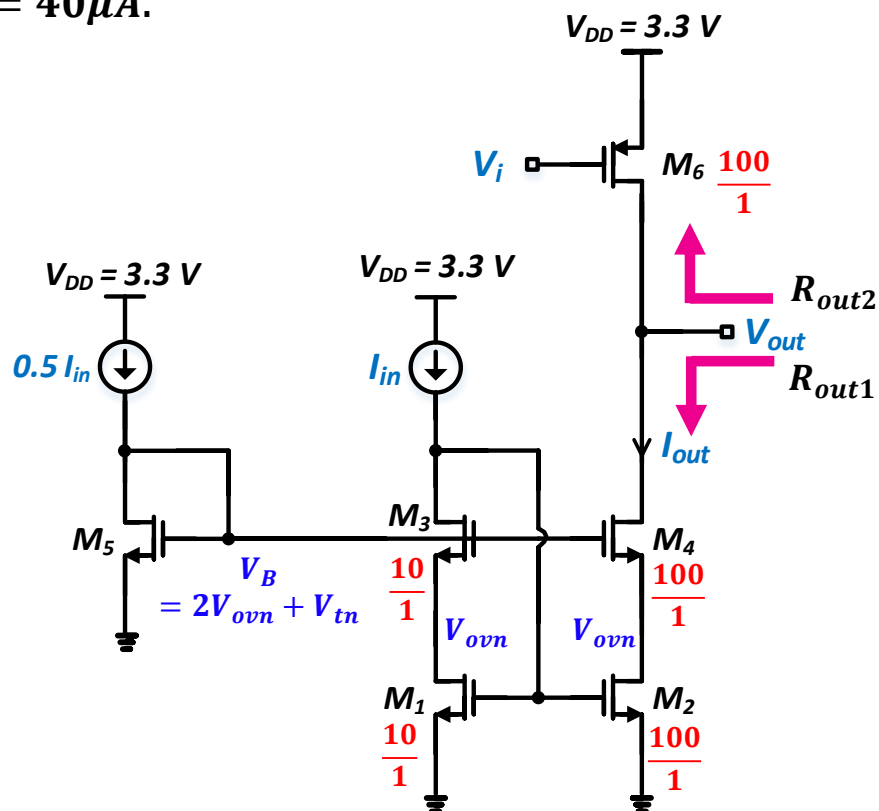
□ $I_{D5} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_5 [V_{GS5} - V_{tn}]^2$

□ $10 \mu A = \frac{1}{2} \times 100 \mu A/V^2 \left(\frac{W}{L}\right)_5 [0.8 - 0.4]^2$

□ $\left(\frac{W}{L}\right)_5 = \frac{5}{4}$

Problem # 8 (2/3)

- The shown amplifier uses the modified cascode current mirror to boost its gain. $\mu_n C_{ox} = 100 \mu A/V^2$, $\mu_p C_{ox} = 50 \mu A/V^2$, $V_{tn} = |V_{tp}| = 0.4 V$, $\lambda_n = |\lambda_p| = 0.1 V^{-1}$. (a) if $I_{in} = 20 \mu A$, find $(W/L)_5$ for maximum output swing, find A_v , $V_{out,min}$, and $V_{out,max}$ (b) Repeat (a) if $I_{in} = 40 \mu A$.



Solution

□ Get A_v

$$\square A_v = G_m \times R_{out} \quad \square G_m = -g_{m6} \quad \square R_{out} = R_{out1} \parallel R_{out2}$$

$$\square A_v = -g_{m6} \times [R_{out1} \parallel R_{out2}]$$

$$\square g_{m6} = \sqrt{2 \mu_p C_{ox} (W/L)_6 \times I_{out}} = 1.4142 \text{ mS}$$

$$\square g_{m4} = \sqrt{2 \mu_n C_{ox} (W/L)_4 \times I_{out}} = 2 \text{ mS}$$

$$\square r_{o2} = r_{o4} = r_o = \frac{1}{\lambda_n I_{out}} = 50 \text{ K}\Omega$$

$$\square r_{o6} = \frac{1}{\lambda_p I_{out}} = 50 \text{ K}\Omega$$

$$\square R_{out1} \approx g_{m4} r_{o4} r_{o2} \approx g_{m4} r_o^2 \approx 5 \text{ M}\Omega$$

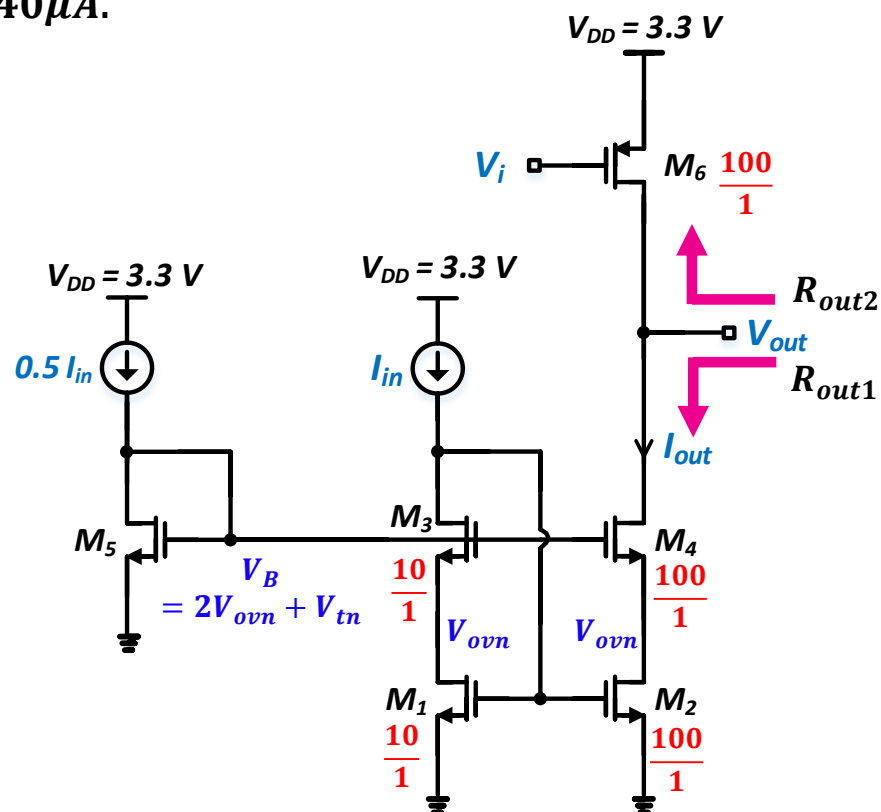
$$\square R_{out2} = r_{o6} = 50 \text{ K}\Omega$$

$$\square R_{out} = R_{out1} \parallel R_{out2} = 49.5 \text{ K}\Omega$$

$$\square A_v = -g_{m6} \times R_{out} = -70 \text{ V/V}$$

Problem # 8 (3/3)

- The shown amplifier uses the modified cascode current mirror to boost its gain. $\mu_n C_{ox} = 100 \mu A/V^2$, $\mu_p C_{ox} = 50 \mu A/V^2$, $V_{tn} = |V_{tp}| = 0.4 V$, $\lambda_n = |\lambda_p| = 0.1 V^{-1}$. (a) if $I_{in} = 20 \mu A$, find $(W/L)_5$ for maximum output swing, find A_v , $V_{out,min}$, and $V_{out,max}$ (b) Repeat (a) if $I_{in} = 40 \mu A$.



Solution

- $$\square \quad V_{out,min} = 2V_{ovn} = 0.4 \text{ V}$$

- $$\blacksquare \quad V_{out,max} = V_{DD} - |V_{ov6}|$$

$$\square \quad |V_{ov6}| = \sqrt{\frac{2 I_{out}}{\mu_p C_{ox} (W/L)_6}} = \sqrt{\frac{2 \times 200 \mu A}{50 \mu A/V^2 \times 100}} = 0.2828 V$$

$$\square \quad V_{out,max} = V_{DD} - |V_{ov6}| = 3.3 \text{ V} - 0.2828 \text{ V} = 3.0172 \text{ V}$$

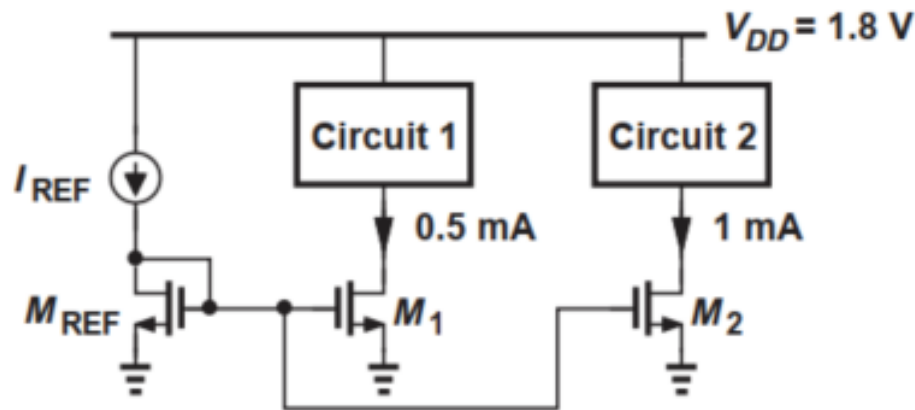
❑ Max output signal swing = $V_{out,max} - V_{out,min} = 3.1072 - 0.4 = 2.3172 \text{ V}$

- ❑ Repeat @ $I_{in} = 40\mu A$

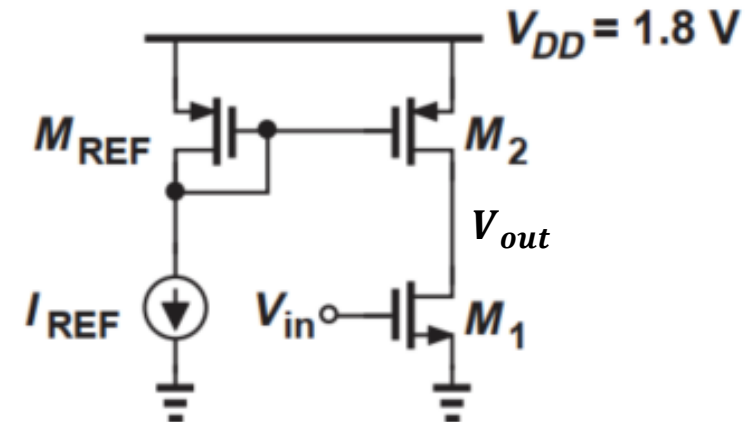
Give your own comments on the obtained results compared with the results you get in (a)

Problem # 9 & 10

- The shown circuit shows an arrangement where M_1 and M_2 serve as current sources for circuits 1 and 2. Design the circuit for a power budget of 3 mW.



- The shown common-source stage must be designed for a voltage gain of 20 V/V and a power budget of 2 mW. Assuming $(W/L)_1 = 20/0.18$, $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox} = 50\text{ }\mu\text{A/V}^2$, $V_{tn} = |V_{tp}| = 0.4\text{ V}$, $\lambda_n = 0.1\text{ V}^{-1}$, and $|\lambda_p| = 0.2\text{ V}^{-1}$, design the circuit.

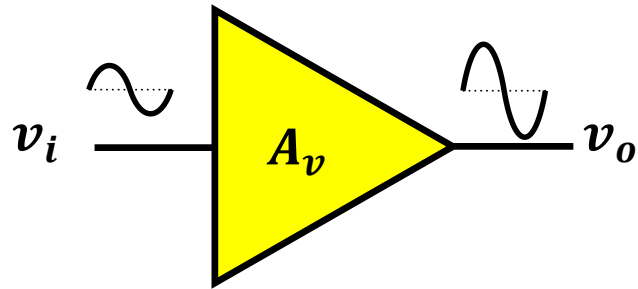


Outline

- ❑ Solved Problems on Current Mirrors
- ❑ **Differential Amplifier**
 - ❑ **Amplifier Types**
 - ❑ Differential Vs Single-ended Signaling
 - ❑ Common-mode and differential signals
 - ❑ BJT Differential Amplifier
 - ❑ Large Signal Operation of BJT Differential Amplifier

Amplifier Types

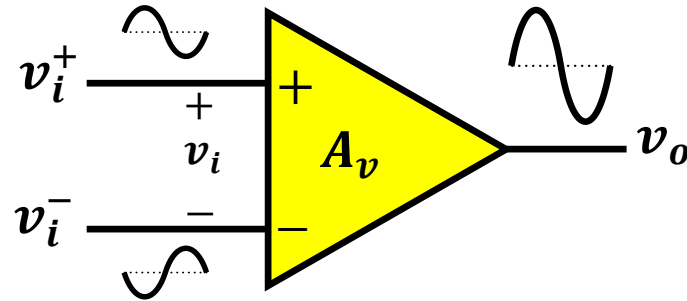
- Single-ended input, single ended output



$$v_o = A_v \times v_i$$

- Input: Single-ended
- Output: Single-ended

- Differential input, single ended output

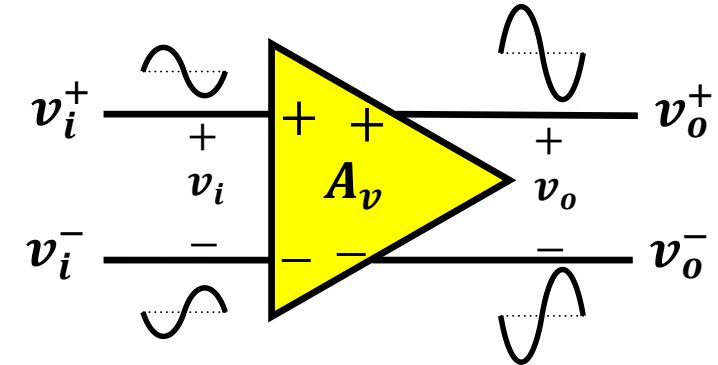


$$v_o = A_v \times v_i$$

$$v_o = A_v \times (v_i^+ - v_i^-)$$

- Input: Differential
- Output: Single-ended

- Differential input, Differential output



$$v_o = A_v \times v_i$$

$$(v_o^+ - v_o^-) = A_v \times (v_i^+ - v_i^-)$$

$$v_o^+ = \frac{A_v}{2} \times (v_i^+ - v_i^-) \quad v_o^- = \frac{-A_v}{2} \times (v_i^+ - v_i^-)$$

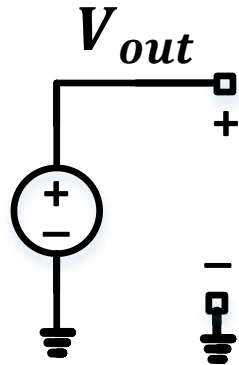
- Input: Differential
- Output: Differential

Outline

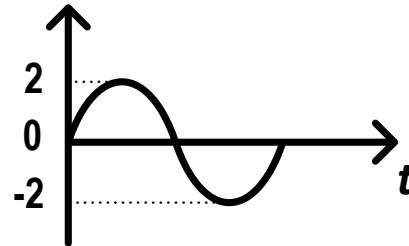
- ❑ Solved Problems on Current Mirrors
- ❑ Differential Amplifier
 - ❑ Amplifier Types
 - ❑ **Differential Vs Single-ended Signaling**
 - ❑ Common-mode and differential signals
 - ❑ BJT Differential Amplifier
 - ❑ Large Signal Operation of BJT Differential Amplifier

Differential Vs Single-Ended Signaling

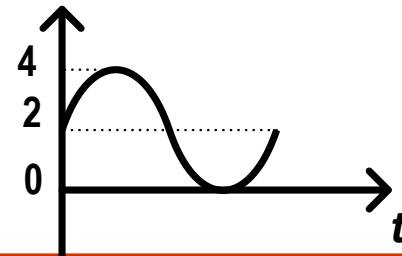
□ **Single-ended (SE):** measured with respect to a fixed reference (usually the ground (0 V))



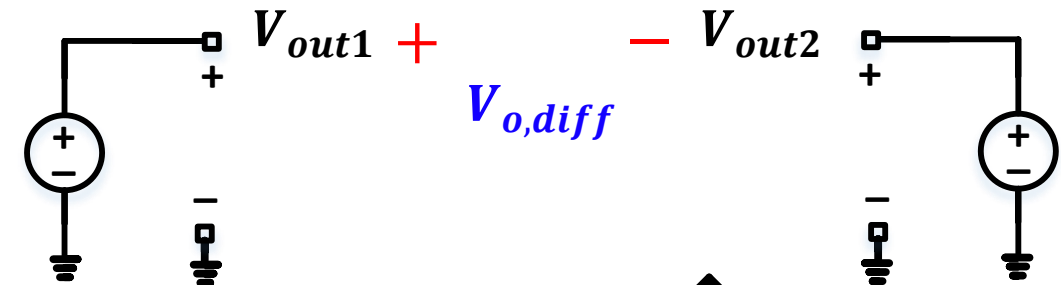
$$V_{out} = 2 \sin \omega t$$



$$V_{out} = 2 + 2 \sin \omega t$$



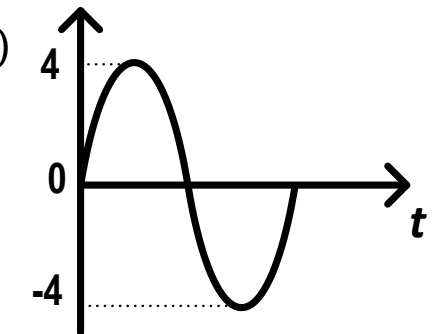
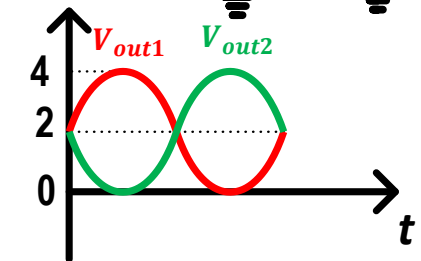
□ **Differential (diff):** measured between two terminals that have equal and opposite amplitudes around a common mode (CM) voltage.



$$V_{out1} = 2 + 2 \sin \omega t$$

$$V_{out2} = 2 - 2 \sin \omega t$$

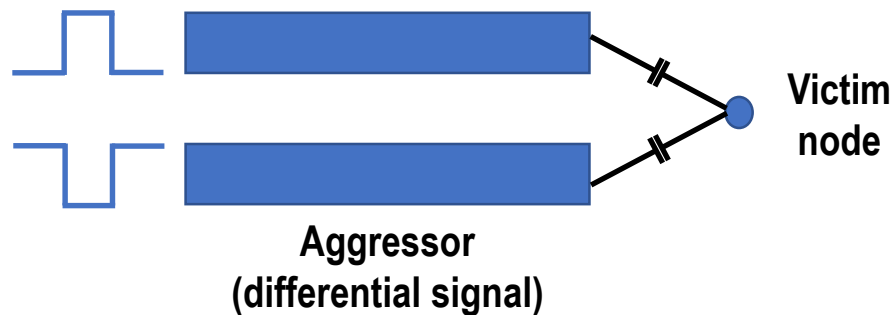
$$\begin{aligned} V_{o,diff} &= V_{out1} - V_{out2} \\ &= (2 + 2 \sin \omega t) - (2 - 2 \sin \omega t) \\ &= 4 \sin \omega t \end{aligned}$$



Differential Vs Single-Ended Signaling

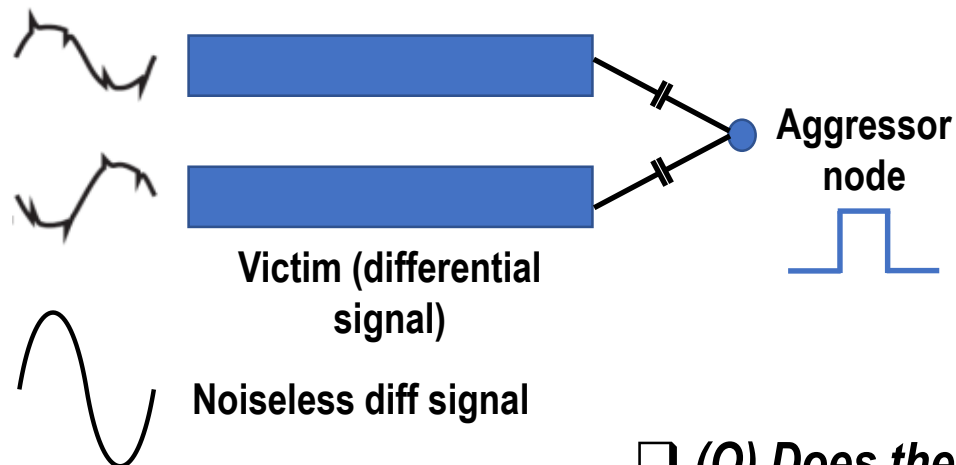
❑ Differential Signaling offers many important benefits.

❑ Reduce Noise coupling to/from other nodes



❑ Assume a fast-switching differential CLK signal exist. (acts as aggressors)

❑ The coupled noises at the victim node are cancelled out



❑ Assume a single-ended CLK signal exist. (acts as aggressor)

❑ The switching noise is coupled equally to each of the single-ended terminals (appears as common-mode noise), consequently the differential output is not affected.

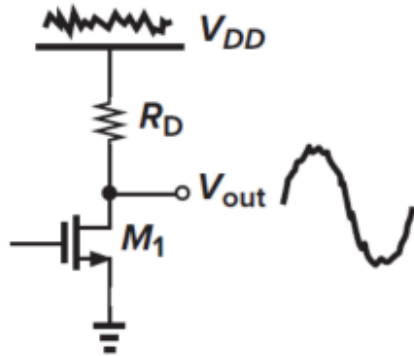
❑ (Q) Does the position of the victim/aggressor node w.r.t differential signal matter?

Differential Vs Single-Ended Signaling

❑ Differential Signaling offers many important benefits.

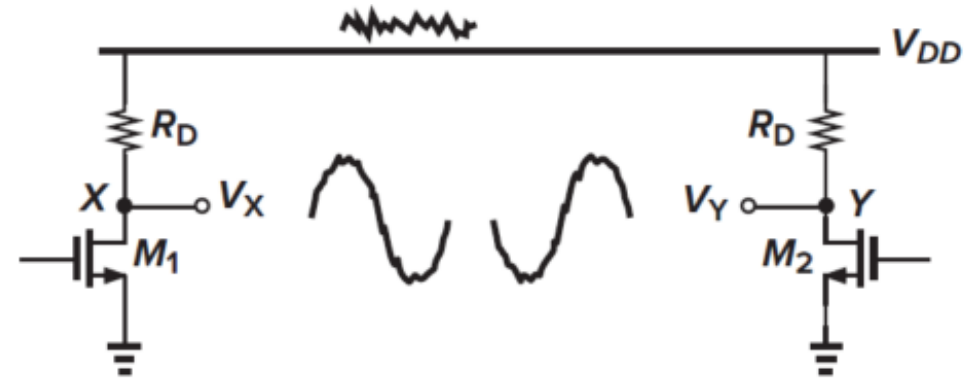
❑ Reduce Noise Coupling from Supply

❑ Single-ended



❑ The supply noise is directly coupled to the single-ended output

❑ Differential



❑ The supply noise is equally directly coupled to the single-ended outputs (V_X and V_Y) (appears as common-mode noise), consequently the differential output is not affected.

Outline

- ❑ Solved Problems on Current Mirrors
- ❑ Differential Amplifier
 - ❑ Amplifier Types
 - ❑ Differential Vs Single-ended Signaling
 - ❑ **Common-mode and differential signals**
 - ❑ BJT Differential Amplifier
 - ❑ Large Signal Operation of BJT Differential Amplifier

Common-mode and differential signals

□ Any two signals (V_1 and V_2) can be expressed in terms of their common-mode signal and differential signal

□ **Common-mode voltage**: it is the average of the two signals

$$\square V_{CM} = \frac{V_1 + V_2}{2}$$

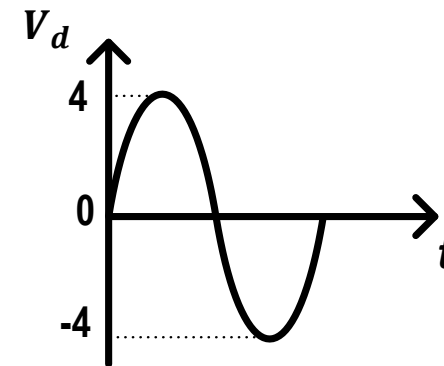
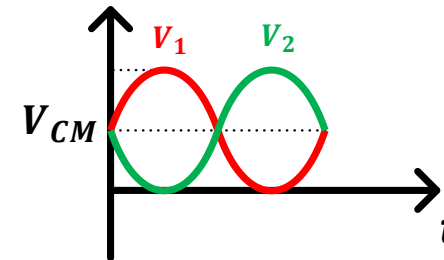
□ **Differential voltage**: it is the average of the two signals

$$\square V_d = V_1 - V_2$$

□ Solve for V_1 and V_2

$$\square V_1 = V_{CM} + \frac{V_d}{2}$$

$$\square V_2 = V_{CM} - \frac{V_d}{2}$$



Common-mode and differential signals

$$\square V_1 = 5V$$

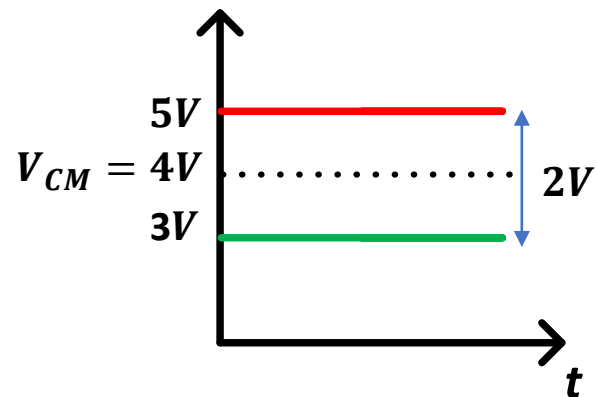
$$\square V_2 = 3V$$

$$\square V_{CM} = \frac{V_1 + V_2}{2} = 4$$

$$\square V_d = V_1 - V_2 = 2$$

$$\square V_1 = V_{CM} + \frac{V_d}{2} = (4) + (1)$$

$$\square V_2 = V_{CM} - \frac{V_d}{2} = (4) - (1)$$



$$\square V_1 = 2V + 2 \sin \omega t$$

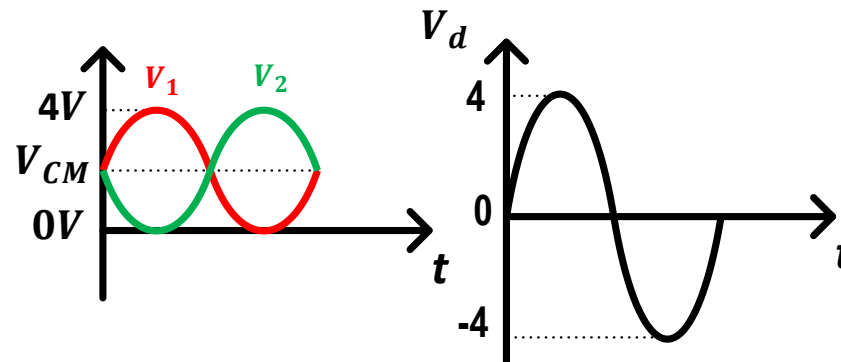
$$\square V_2 = 2V - 2 \sin \omega t$$

$$\square V_{CM} = \frac{V_1 + V_2}{2} = 2V$$

$$\square V_d = V_1 - V_2 = 4 \sin \omega t$$

$$\square V_1 = V_{CM} + \frac{V_d}{2} = (2) + (2 \sin \omega t)$$

$$\square V_2 = V_{CM} - \frac{V_d}{2} = (2) - (2 \sin \omega t)$$



$$\square V_1 = 3V + 3 \sin \omega t$$

$$\square V_2 = 1V - 1 \sin \omega t$$

$$\square V_{CM} = \frac{V_1 + V_2}{2} = 2V + 1 \sin \omega t$$

$$\square V_d = V_1 - V_2 = 2V + 4 \sin \omega t$$

$$\square V_1 = V_{CM} + \frac{V_d}{2} = (2V + 1 \sin \omega t) + (1V + 2 \sin \omega t)$$

$$\square V_2 = V_{CM} - \frac{V_d}{2} = (2V + 1 \sin \omega t) - (1V + 2 \sin \omega t)$$

$\square$ Plot V_1 , V_2 , V_{CM} , and V_d

Outline

- ❑ Solved Problems on Current Mirrors
- ❑ Differential Amplifier
 - ❑ Amplifier Types
 - ❑ Differential Vs Single-ended Signaling
 - ❑ Common-mode and differential signals
 - ❑ **BJT Differential Amplifier**
 - ❑ Large Signal Operation of BJT Differential Amplifier

BJT Differential Amplifier

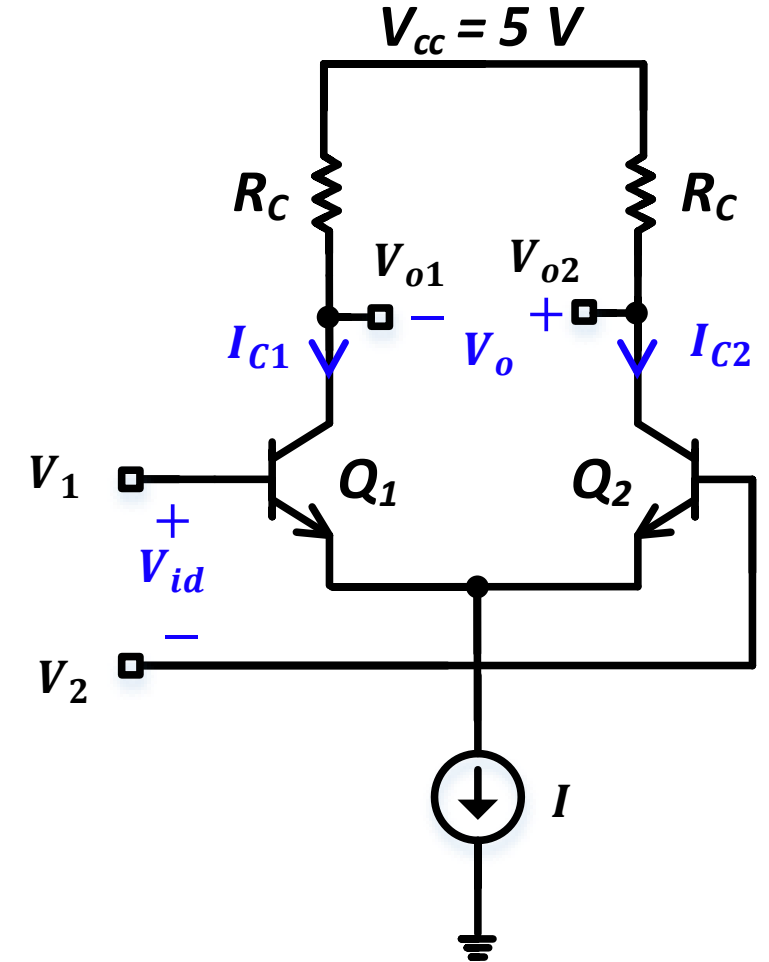
❑ The shown circuit shows a simple BJT differential amplifier with resistive load (R_C)

❑ Q_1 and Q_2 are **matched** (same I_S and β). Q_1 and Q_2 are called differential pair.

❑ The total drawn current from V_{cc} is **constant** and equals the tail current source (I), regardless of the input voltage

❑ According to the input voltage (V_1 and V_2), I_{C1} and I_{C2} will be determined, and consequently the output voltage.

❑ The tail current source (I) is implemented using any of the current mirror circuits discussed in the previous chapter.



Common-Mode operation

□ If $V_1 = V_2 = V_{CM}$, find V_o

□ $V_{id} = V_1 - V_2 = 0$

□ As Q_1 and Q_2 are matched

$$\square V_E = V_{CM} - V_{BE} = V_{CM} - 0.7 \text{ V}$$

$$\square I_{C1} = I_{C2} = \frac{I}{2}$$

$$\square V_{o1} = V_{CC} - I_{C1}R_C = V_{CC} - \left(\frac{I}{2}\right)R_C$$

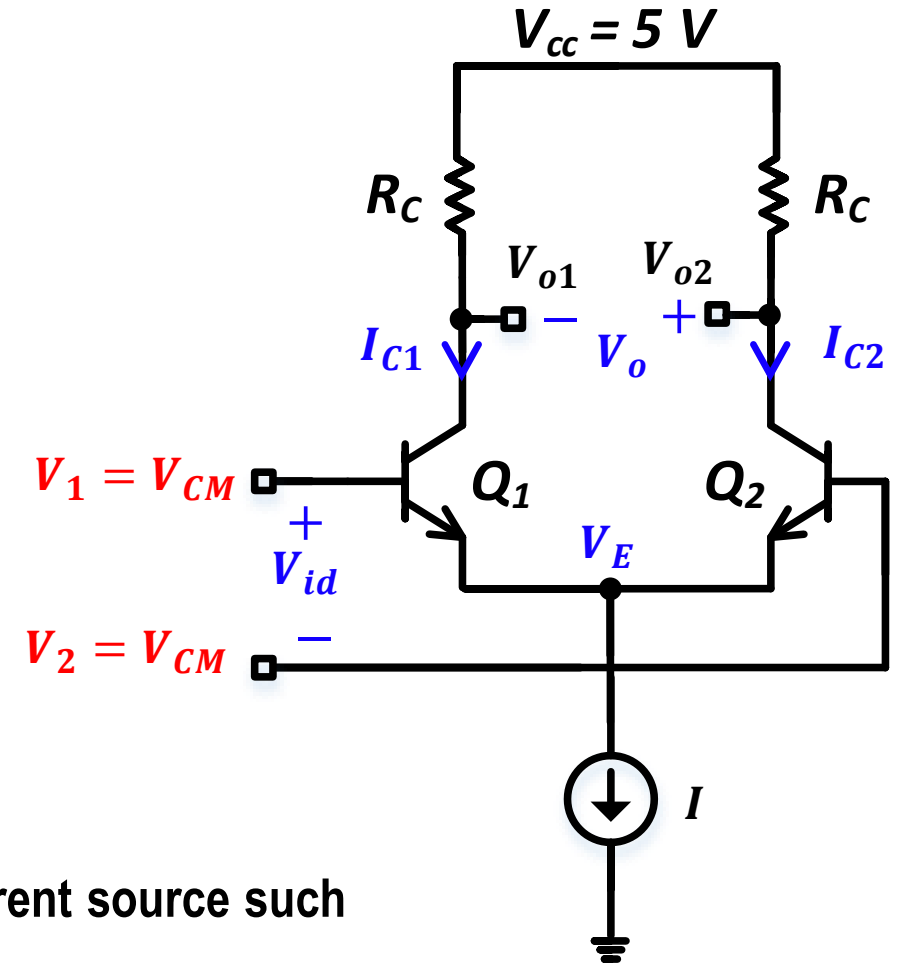
$$\square V_{o2} = V_{CC} - I_{C2}R_C = V_{CC} - \left(\frac{I}{2}\right)R_C$$

$$\square V_o = V_{o2} - V_{o1} = 0$$

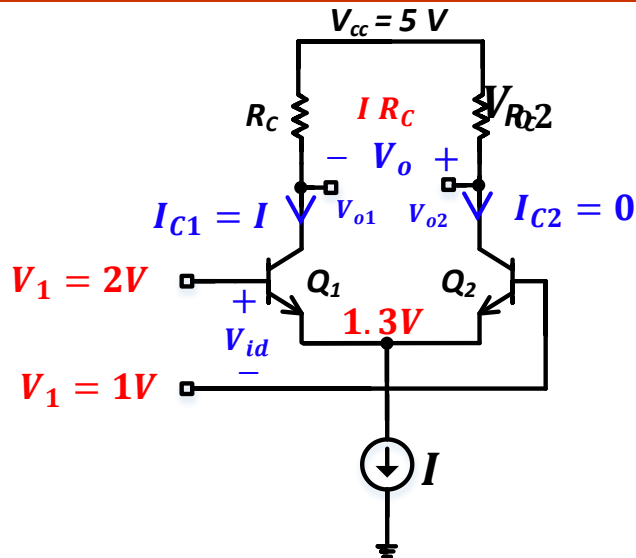
□ The differential amplifier does not response to common mode input voltage and the output voltage equal zero

□ Note: V_{CM} should be grater than 0.7 V + min required voltage of current source such that the tail current remains constant.

□ If $V_1 = V_2 = 0.5 \text{ V} \rightarrow Q1$ and $Q2$ are off and tail current drops to zero $\rightarrow I_{C1} = I_{C2} = 0 \rightarrow V_{o1} = V_{o2} = V_{CC} \rightarrow V_o = 0 \text{ V}$

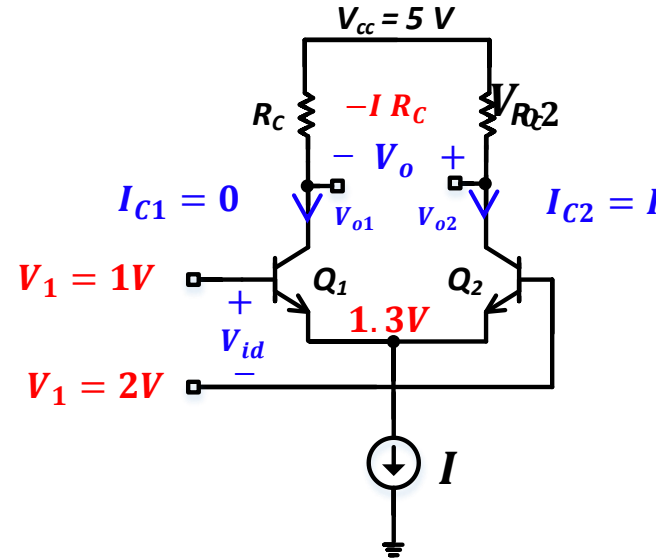


Differential Mode operation

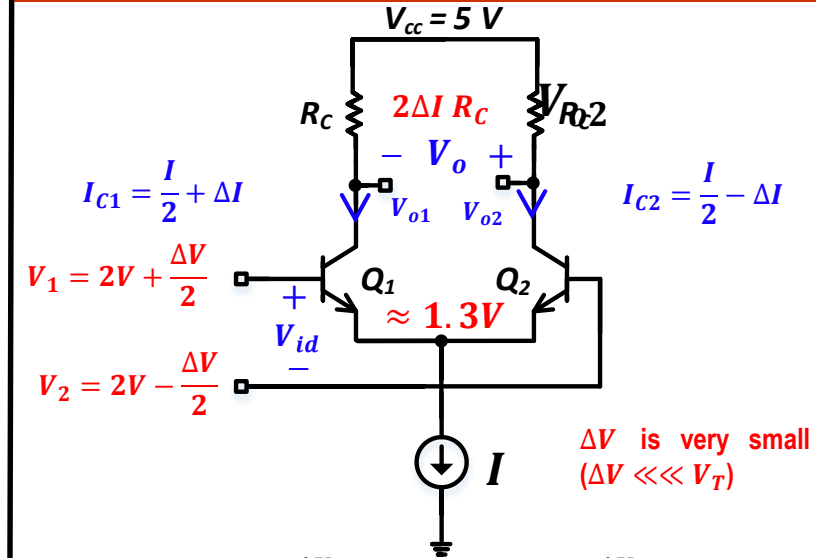


- If $V_1 = 2V$ and $V_2 = 1V$ □ $V_{id} = 1V$
- $Q1 \rightarrow \text{ON}, Q2 \rightarrow \text{OFF}$ □ $V_E = 1.3V$
- $V_{BE1} = 0.7V$ □ $V_{BE2} = -0.3V$
- $I_{C1} = I$ □ $I_{C2} = 0$
- $V_{o1} = V_{CC} - IR_C$ □ $V_{o2} = V_{CC}$
- $V_o = V_{o2} - V_{o1} = IR_C$

□ The differential amplifier responds to the differential input voltage and the output voltage depends on differential input



- If $V_1 = 1V$ and $V_2 = 2V$ □ $V_{id} = -1V$
- $Q1 \rightarrow \text{OFF}, Q2 \rightarrow \text{ON}$ □ $V_E = 1.3V$
- $V_{BE1} = -0.3V$ □ $V_{BE2} = 1.3V$
- $I_{C1} = 0$ □ $I_{C2} = I$
- $V_{o1} = V_{CC}$ □ $V_{o2} = V_{CC} - IR_C$
- $V_o = V_{o2} - V_{o1} = -IR_C$



- If $V_1 = 2V + \frac{\Delta V}{2}$ and $V_2 = 2V - \frac{\Delta V}{2}$
- $V_{id} = \Delta V$
- $Q1 \rightarrow \text{ON}, Q2 \rightarrow \text{ON}$
- $V_{BE1} = 0.7V + \frac{\Delta V}{2}$ □ $V_{BE2} = 0.7V - \frac{\Delta V}{2}$
- $I_{C1} = \frac{I}{2} + \Delta I$ □ $I_{C2} = \frac{I}{2} - \Delta I$
- $V_{o1} = V_{CC} - \left(\frac{I}{2} + \Delta I\right) R_C$
- $V_{o2} = V_{CC} - \left(\frac{I}{2} - \Delta I\right) R_C$
- $V_o = V_{o2} - V_{o1} = 2\Delta I R_C$

Outline

- ❑ Solved Problems on Current Mirrors
- ❑ Differential Amplifier
 - ❑ Amplifier Types
 - ❑ Differential Vs Single-ended Signaling
 - ❑ Common-mode and differential signals
 - ❑ BJT Differential Amplifier
 - ❑ **Large Signal Operation of BJT Differential Amplifier**

Large Signal Operation

□ Find I_{C1} , I_{C2} , and V_o in terms of V_{id}

Solution

□ $I_{C1} = I_S e^{(V_{BE1}/V_T)}$ (1)

□ $I_{C2} = I_S e^{(V_{BE2}/V_T)}$ (2)

□ $V_{BE1} = V_1 - V_E$

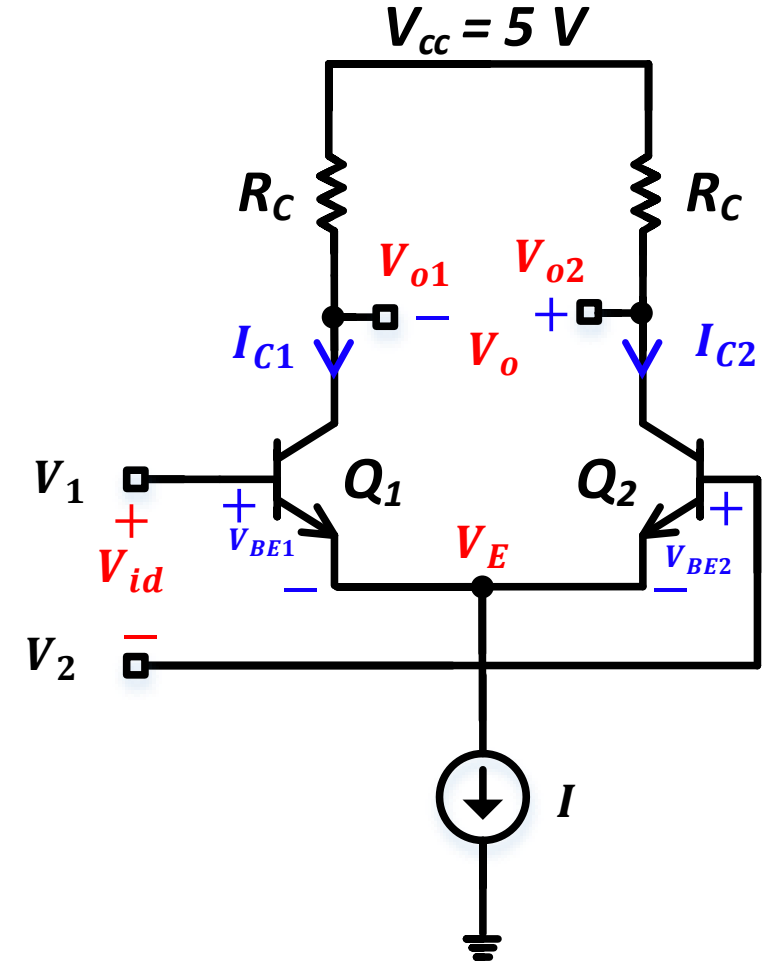
□ $V_{BE2} = V_2 - V_E$

□ $V_{id} = V_1 - V_2 = V_{BE1} - V_{BE2}$

□ Assume $\alpha \approx 1 \rightarrow I = I_{C1} + I_{C2}$ & use (1) and (2)

□ $I = I_{C1} \left[1 + \frac{I_{C2}}{I_{C1}} \right] = I_{C1} [1 + e^{([V_{BE2} - V_{BE1}]/V_T)}] = I_{C1} [1 + e^{(-V_{id}/V_T)}]$

□ $I = I_{C2} \left[1 + \frac{I_{C1}}{I_{C2}} \right] = I_{C2} [1 + e^{([V_{BE1} - V_{BE2}]/V_T)}] = I_{C2} [1 + e^{(+V_{id}/V_T)}]$



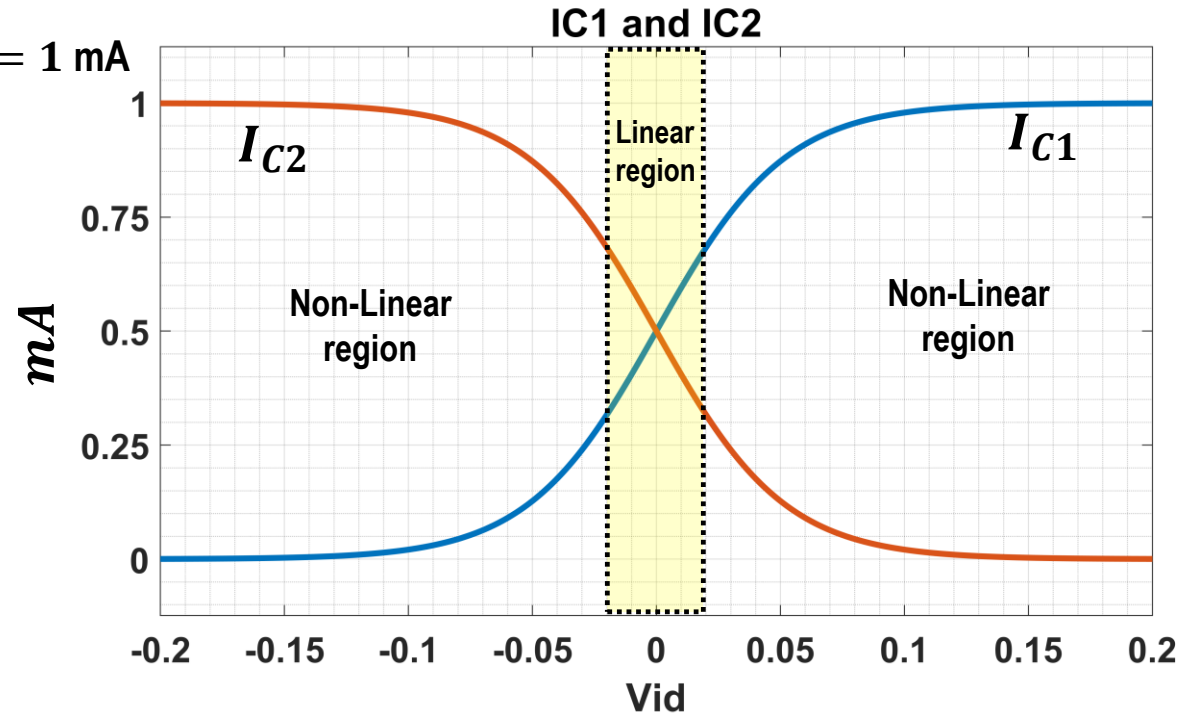
□ Q_1 and Q_2 are matched

Large Signal Operation

$$\square I_{C1} = \frac{I}{1+e^{(-V_{id}/V_T)}}$$

$$\square I_{C2} = \frac{I}{1+e^{(+V_{id}/V_T)}}$$

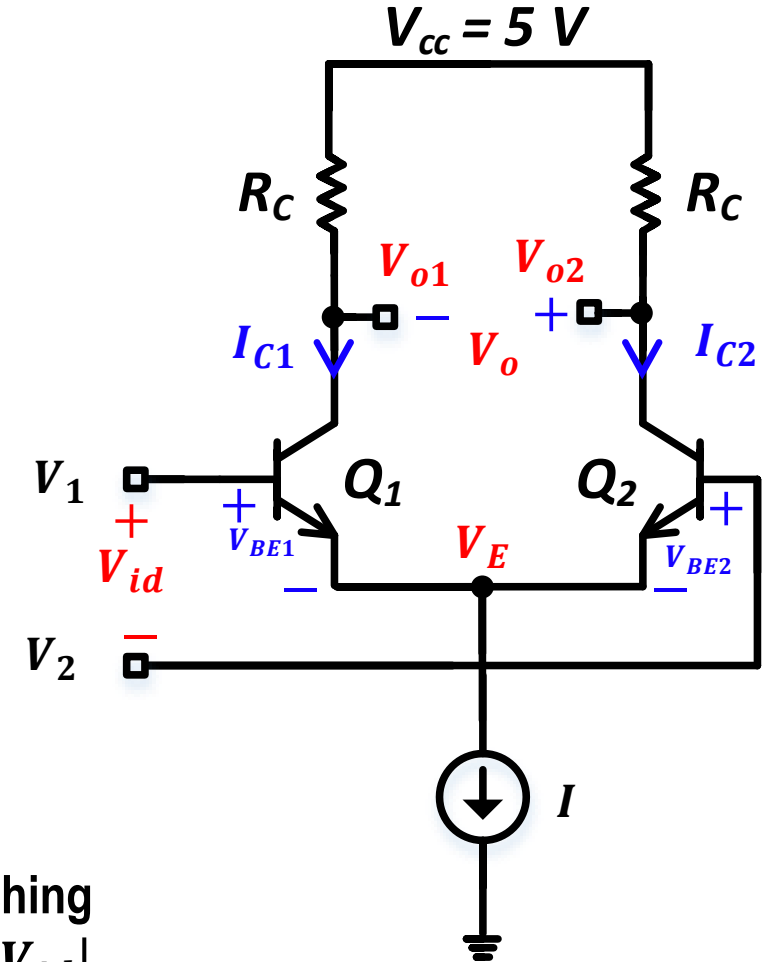
□ At $I = 1 \text{ mA}$



□ Linear region: $V_{id} \ll 2V_T$

□ At $V_{id} = 0 \rightarrow I_{C1} = I_{C2} = \frac{I}{2}$

□ Complete current switching approximately occurs at $|V_{id}| \approx 4V_T \approx 0.104 \text{ V}$



Large Signal Operation

$$\square I_{C1} = \frac{I}{1+e^{(-V_{id}/V_T)}}$$

$$\square I_{C2} = \frac{I}{1+e^{(+V_{id}/V_T)}}$$

$$\square V_o = V_{o2} - V_{o1} = [V_{CC} - I_{C2}R_C] - [V_{CC} - I_{C1}R_C]$$

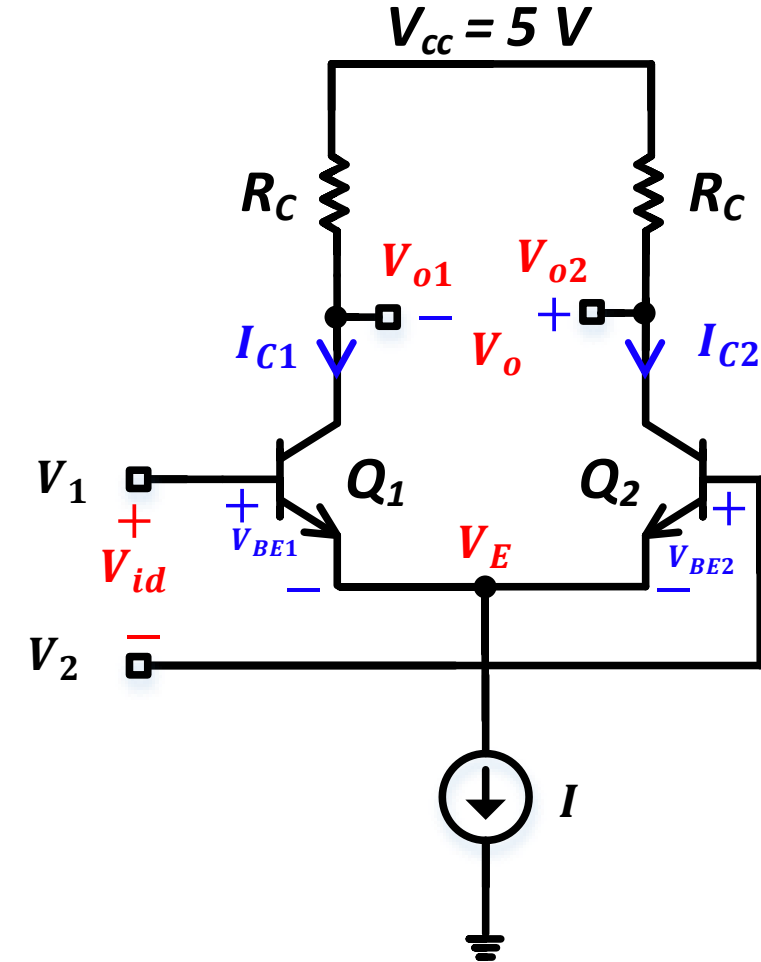
$$\square V_o = V_{o2} - V_{o1} = [I_{C1} - I_{C2}] R_C$$

$$\square V_o = \left[\frac{I}{1+e^{(-V_{id}/V_T)}} - \frac{I}{1+e^{(+V_{id}/V_T)}} \right] R_C$$

$$\square V_o = I R_C \left[\frac{e^{(+V_{id}/2V_T)}}{e^{(+V_{id}/2V_T)} + e^{(-V_{id}/2V_T)}} - \frac{e^{(-V_{id}/2V_T)}}{e^{(-V_{id}/2V_T)} + e^{(+V_{id}/2V_T)}} \right]$$

$$\square V_o = I R_C \left[\frac{e^{(+V_{id}/2V_T)} - e^{(-V_{id}/2V_T)}}{e^{(+V_{id}/2V_T)} + e^{(-V_{id}/2V_T)}} \right]$$

$$\square V_o = I R_C \times \tanh \left(\frac{V_{id}}{2V_T} \right)$$



$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Large Signal Operation

$$\square V_o = I R_C \times \tanh\left(\frac{V_{id}}{2V_T}\right)$$

$$\square \text{ At } I = 1 \text{ mA}, R_C = 1 \text{ K}\Omega$$

$$\square \text{ At } V_{id} = 0$$

$$\square I_{C1} = I_{C2} = 0.5 \text{ mA}$$

$$\square V_{o1} = V_{o2} = (5 \text{ V}) - (0.5 \text{ mA} \times 1 \text{ K}) = 4.5 \text{ V}$$

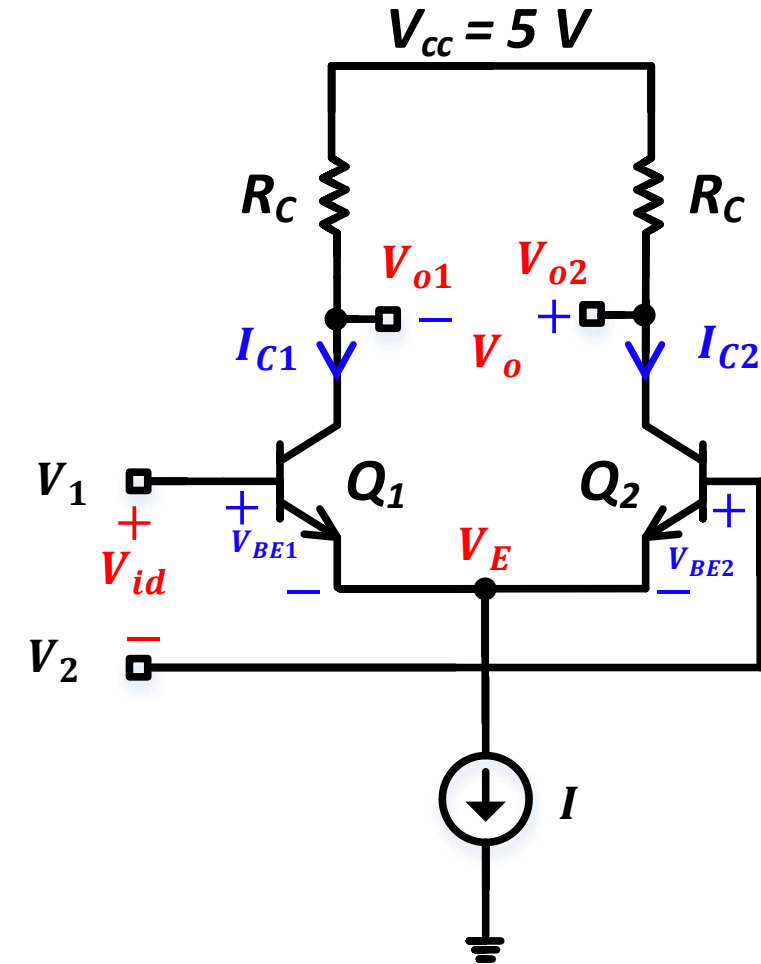
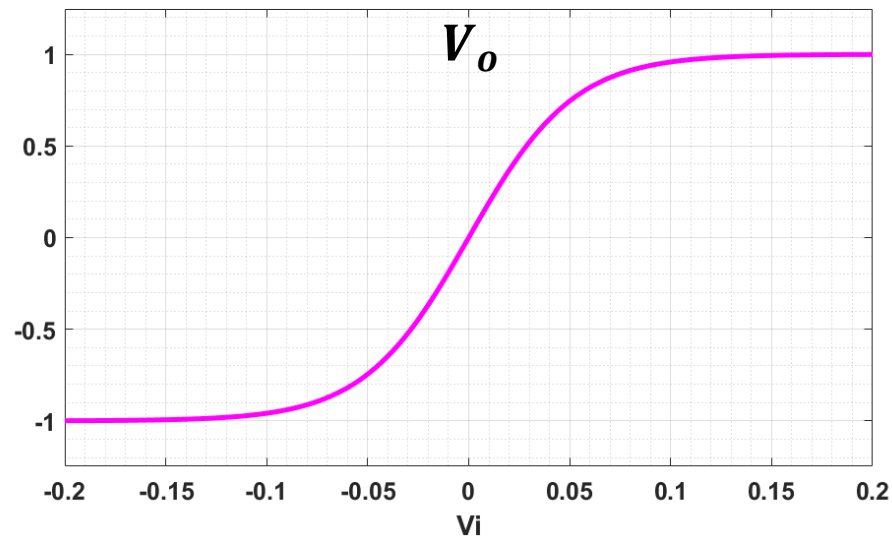
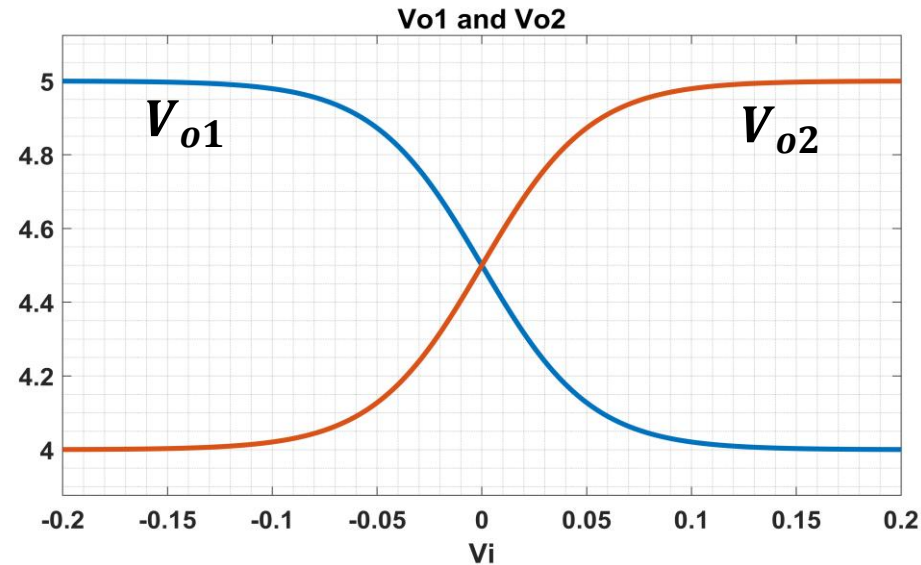
$$\square V_o = 0 \text{ V}$$

$$\square \text{ At } V_{id} = \text{Large +ve voltage}$$

$$\square I_{C1} = 1 \text{ mA} \quad \square I_{C2} = 0$$

$$\square V_{o1} = 4 \text{ V} \quad \square V_{o2} = 5 \text{ V}$$

$$\square V_o = 1 \text{ V}$$



Large Signal Operation

$$\square V_o = I R_C \times \tanh \left(\frac{V_{id}}{2V_T} \right)$$

$$\square \tanh(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\square \text{At } x \lll 1, \tanh(x) = x$$

$$\square \text{In linear region with } V_{id} \lll 2V_T$$

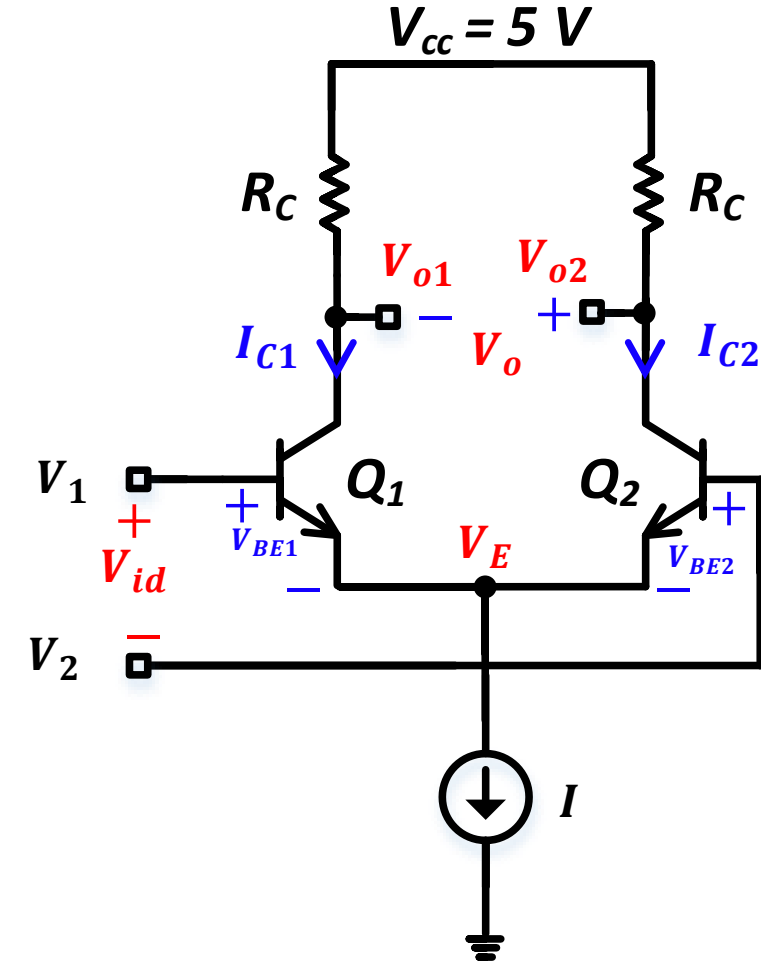
$$\square V_o = I R_C \times \left(\frac{V_{id}}{2V_T} \right)$$

$$\square V_o = \frac{I/2}{V_T} R_C \times V_{id}$$

$$\square V_o = g_m R_C \times V_{id}$$

$$\square \frac{V_o}{V_{id}} = g_m R_C$$

$$\square \text{Note: } g_m = \frac{I_C}{V_T} = \frac{I/2}{V_T}$$



References

- Gray, Paul R., et al, “*Analysis and Design of Analog Integrated Circuits*”, 5th edition, John Wiley & Sons, 2009.
- Behzad Razavi, “*Fundamentals of Microelectronics*”, 2th edition, Wiley, 2014.
- Behzad Razavi, “*Design of Analog CMOS Integrated Circuits*”, 2nd edition, MC Graw Hill, 2016.

□ Thanks